PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T58 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

9 classified items • 36 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T58 • Expertise Q16+ • 9 items • 36 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Combined & Conditional Probability

Unit 1 • Expertise • 36 marks • 9 item(s)

June 2025 | 4MA1-1H Q21 | 3 marks

Calculate an exact composition in three or more draws

June 2025 | 4MA1-2H Q20 | 6 marks

Calculate a direct given-that probability

June 2024 | 4MA1-1H Q16 | 4 marks

Use a complement in repeated no-replacement selection

November 2023 | 4MA1-1H Q20 | 4 marks

Find exactly one, same or different independent outcomes

June 2022 | 4MA1-2H Q16 | 8 marks

Calculate a direct given-that probability

January 2020 | 4MA1-2H Q18 | 2 marks

Use a histogram or table as a conditional sample space

November 2021 | 4MA1-1H Q16 | 1 marks

Calculate a direct given-that probability

November 2021 | 4MA1-2H Q16 | 3 marks

Compare with-replacement and without-replacement models

May-Nov 2020 | 2H Q20 | 5 marks

Recover an unknown population from no-replacement probability

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q21 **3 marks**

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 21 is 3 marks)

WORKING SPACE

4MA1-1H Q21

3 marks

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 21 is 3 marks)

Exactly two counters of the same colour **3 marks**

Use the complement

$$P(\text{exactly two}) = 1 - P(\text{all same}) - P(\text{all different}).$$

All same

$$P(\text{all same}) = \frac{9 \cdot 8 \cdot 7 + 7 \cdot 6 \cdot 5 + 4 \cdot 3 \cdot 2}{20 \cdot 19 \cdot 18} = \frac{738}{6840}.$$

All different

$$P(\text{all different}) = \frac{6(9 \cdot 7 \cdot 4)}{20 \cdot 19 \cdot 18} = \frac{1512}{6840}.$$

Finish

$$1 - \frac{738 + 1512}{6840} = \frac{4590}{6840} = \frac{51}{76}.$$

Final answer

$$\frac{51}{76}$$

Dependency check Sampling is without replacement; the factor 6 counts the orders of three different colours.

4MA1-2H Q20 **6 marks**

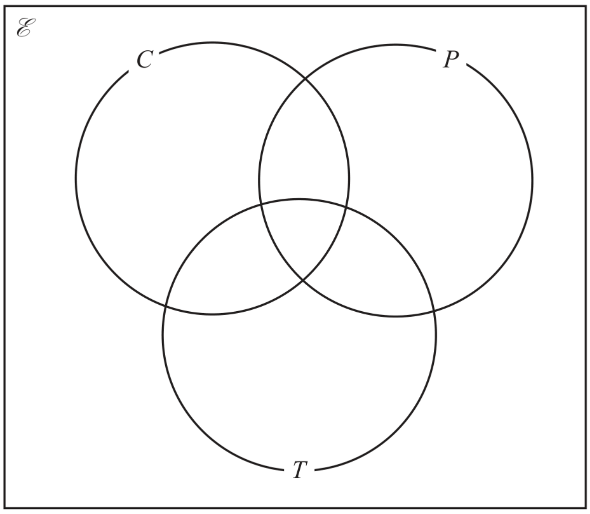
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....
(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....
(2)

WORKING SPACE

4MA1-2H Q20 **6 marks**

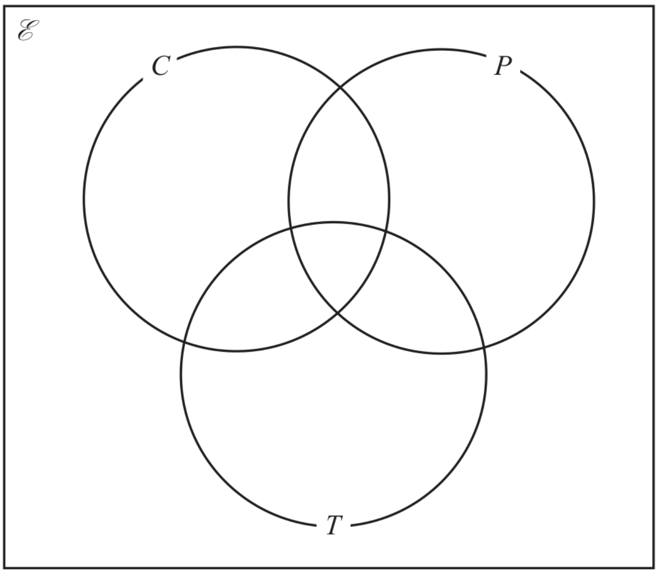
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....

(2)

Three-set Venn diagram and conditional probability **6 marks**

Intersections

$$C\cap P\cap T=12,\ CP \text{ only}=6,\ CT \text{ only}=15,\ PT \text{ only}=20.$$

Single-set regions

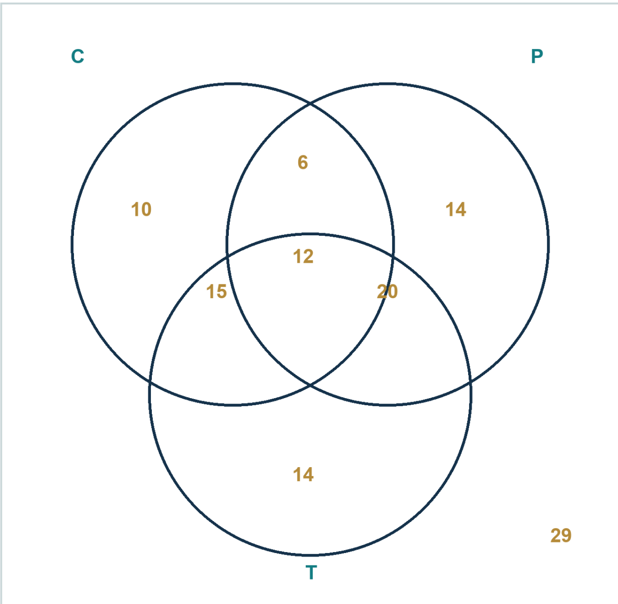
C only $=43-(6+15+12)=10$. If P only $=T$ only $=k$, $10+6+15+12+20+2k=120-29,$ so $k=14$.

Part (b)

$$n(T'\cap C)=10+6=16.$$

Part (c)

$$P(P\mid C)=\frac{n(P\cap C)}{n(C)}=\frac{18}{43}.$$



Solved reference diagram

Final answer

$$\text{Regions } (10, 14, 14, 6, 15, 20, 12;\ \text{outside } 29),\quad 16,\quad \frac{18}{43}$$

Dependency check Parts (b) and (c) read the completed diagram; the diagram appears once.

4MA1-1H Q16 **4 marks**

16 There are 20 sweets in a box.

15 of the sweets are red
5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

....

WORKING SPACE

4MA1-1H Q16

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red
5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

....

At least one of each colour **4 marks**

Complement

$$1 - \frac{15}{20}\frac{14}{19}\frac{13}{18} - \frac{5}{20}\frac{4}{19}\frac{3}{18} = \frac{45}{76}.$$

Final answer

$$\frac{45}{76}$$

4MA1-1H Q20 demand **4 marks**

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

....

WORKING SPACE

4MA1-1H Q20 demand **4 marks**

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

....

Probability of mean 15 cents **4 marks**

Demand – Probability of mean 15 cents

Find each coin probability

$$P(10) = \sqrt{49/121} = 7/11, \quad P(20) = 4/11.$$

Use the two orders

$$P(10, 20 \text{ or } 20, 10) = 2 \times \frac{7}{11} \times \frac{4}{11} = \frac{56}{121}.$$

Final answer

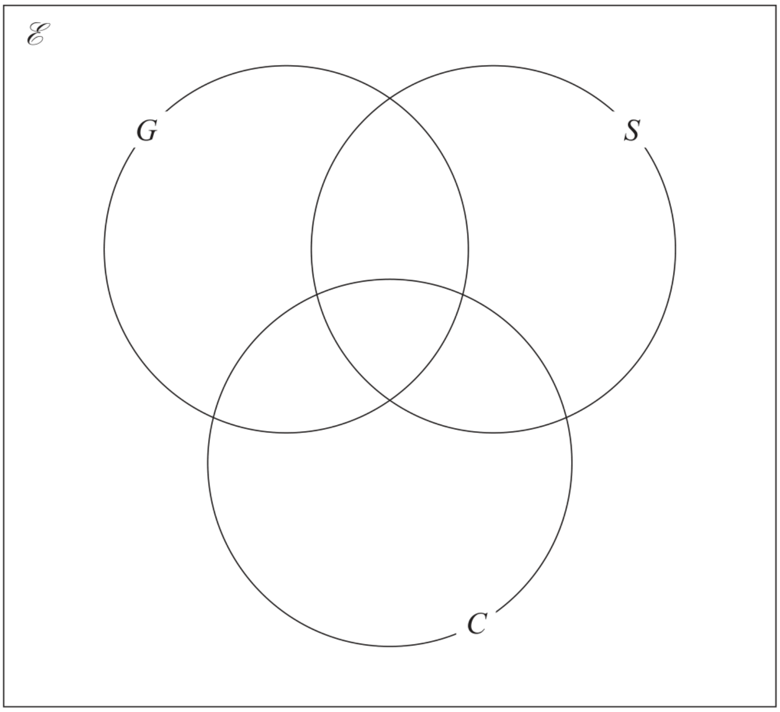
$56/121$

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c) **8 marks**

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers
- 31 have sheep
- 53 have chickens
- 6 have goats, sheep and chickens
- 11 have sheep and goats
- 17 have sheep and chickens
- 18 have goats and chickens
- 20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c) **8 marks**

(i) $n(G)$

....
(1)

(ii) $n([G \cup S]')$

....
(1)

(iii) $n(G' \cap C)$

....
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

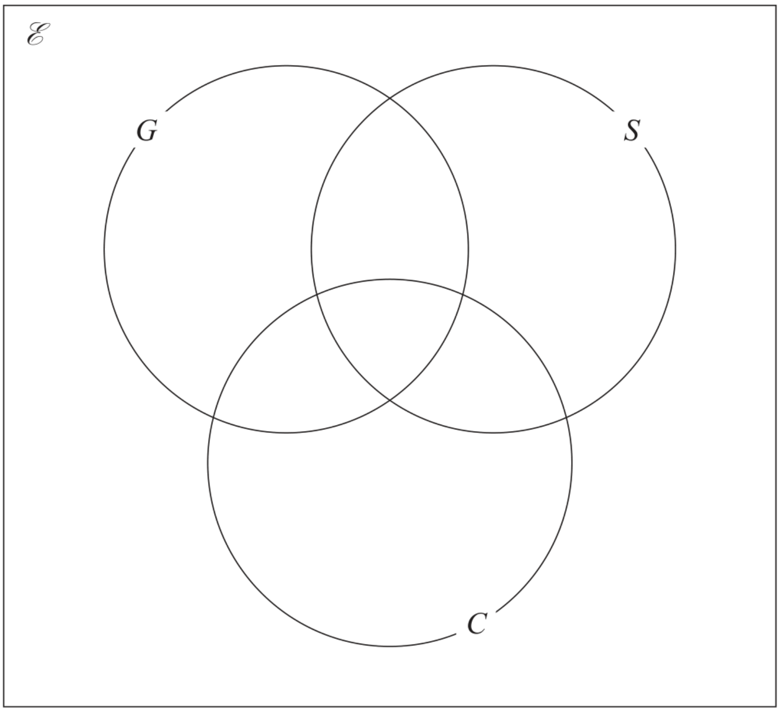
....
(2)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c) **8 marks**

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers
- 31 have sheep
- 53 have chickens
- 6 have goats, sheep and chickens
- 11 have sheep and goats
- 17 have sheep and chickens
- 18 have goats and chickens
- 20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c) **8 marks**

(i) $n(G)$

....

(1)

(ii) $n([G \cup S]')$

....

(1)

(iii) $n(G' \cap C)$

....

(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

....

(2)

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability **8 marks**

Q16(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Complete a three-set Venn diagram

Method
Put 6 in the triple intersection; then GS-only = 11 - 6 = 5, SC-only = 17 - 6 = 11, and GC-only = 18 - 6 = 12.
Why: The pair totals include the triple intersection.

Method
S-only = 31 - (5 + 11 + 6) = 9 and C-only = 53 - (11 + 12 + 6) = 24.
Why: Subtract every already-filled sheep or chicken region.

Method
There are 80 farmers inside the three sets, so G-only = 80 - (9 + 24 + 5 + 11 + 12 + 6) = 13; put 20 outside.
Why: Use the universal total and the 20 with none.

Final answer
Venn regions: G only 13; S only 9; C only 24; GS only 5; GC only 12; SC only 11; GSC 6; outside 20

Q16(b)(i) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method
n(G) = 13 + 5 + 12 + 6 = 36.
Why: Add every region inside G.

Final answer
n(G): 36

Q16(b)(ii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method
(G union S)′ contains C-only and the outside region, so n((G union S)′) = 24 + 20 = 44.
Why: The complement excludes every point in G or S.

Final answer
count: 44

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method
G′ intersection C contains C-only and SC-only, so n(G′ intersection C) = 24 + 11 = 35.
Why: Include chicken regions that are outside G.

Final answer
count: 35

Q16(c) – Chapter: Combined & Conditional Probability

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method
The chosen farmer comes from the 53 farmers with chickens.
Why: This is the conditional sample space.

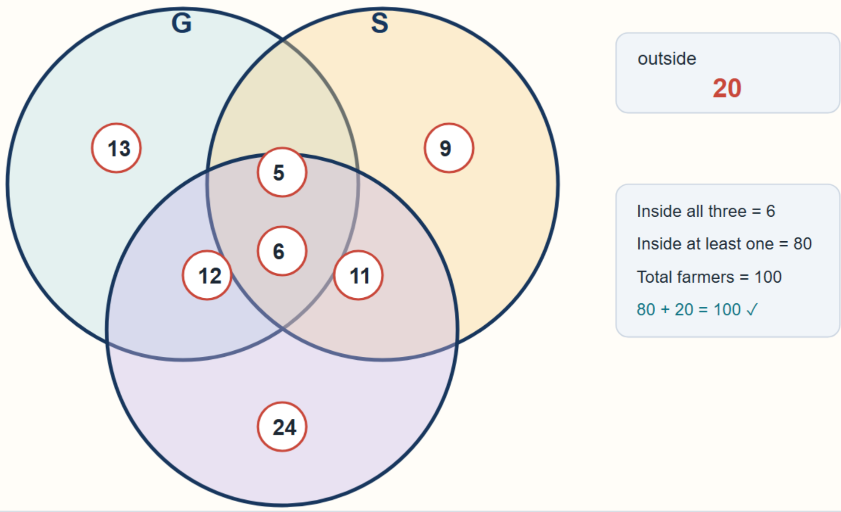
Method
Of these, 18 also have goats, so the probability is 18/53.
Why: Use the G-and-C count over the C count.

Final answer
probability: 18/53

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability **8 marks**

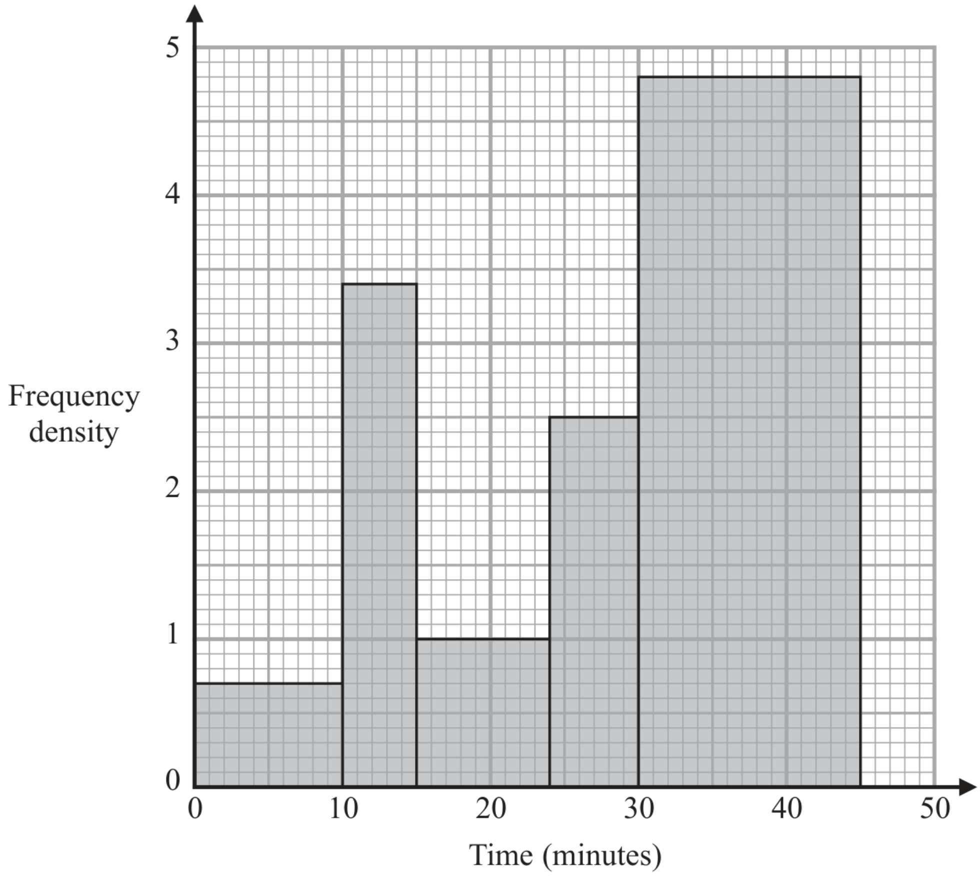
Q16(a) Completed three-set Venn diagram

Place every region, including the outside count



Elite IGCSE Mathematics • Dr Eslam Ahmed • eliteigcse.com

18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

(b) find the probability that this customer spent more than 36 minutes in the supermarket.

....

(2)

Worked solution

January 2020 | Question 18 | 2 marks

Family: Use conditional probability and restricted sample spaces • Variant: Use a histogram or table as a conditional sample space

Method / working

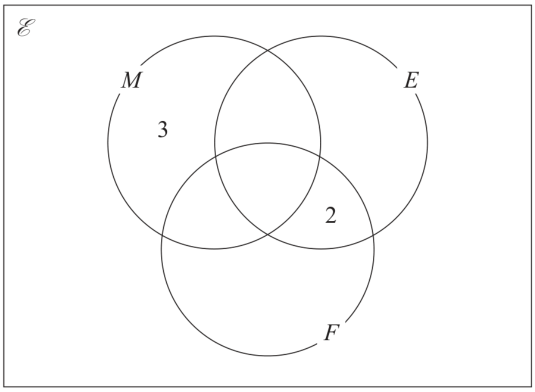
1. Given more than 30 minutes, use the 30-45 class.
2. The part above 36 is 9 of the 15 minutes, so probability=9/15=3/5.

Final answer: 3/5

16 There are 32 students in a class.
In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).

25 students passed the test in Maths.
20 students passed the test in English.
14 students passed the test in French.
18 students passed the tests in Maths and English.
11 students passed the tests in Maths and French.
4 students failed all three tests.
x students passed all three tests.

The incomplete Venn diagram gives some more information about the results of the 32 students.

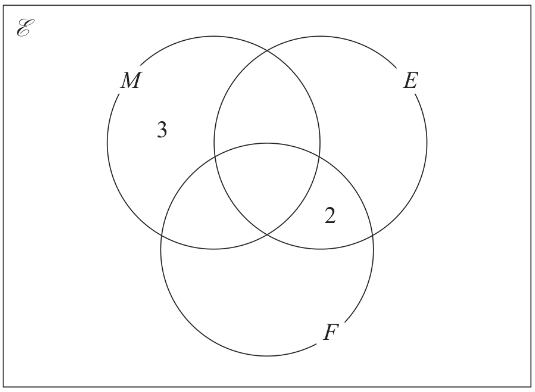


(a) Use all the given information about the results of students who passed the test in Maths to find the value of x.

x = ...

(2)

(b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

(c) Find the probability that this student failed the test in French.

....

(1)

Worked solution

November 2021 | Question 16 | 1 marks

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method / working

1. Use the Maths circle.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. M only$=3$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. $M\cap E$ only$=18-x$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $M\cap F$ only$=11-x$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. The total in M is 25:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. $3+(18-x)+(11-x)+x=25$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. $32-x=25$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. $x=7$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. So:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. $M\cap E$ only$=11,\quad M\cap F$ only$=4,\quad E\cap F$ only$=2$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. E only$=20-11-2-7=0$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. F only$=14-4-2-7=1$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. For a student who passed Maths, failed French means M but not F:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
14. $\frac{3+11}{25}=\frac{14}{25}$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
15. Answers: $x=7$, and $\frac{14}{25}$.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: $x=7$, and $\frac{14}{25}$.

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

....

Worked solution

November 2021 | Question 16 | 3 marks

Family: Combine set counts with sequential probability • Variant: Compare with-replacement and without-replacement models

Method / working
1. There are \(4\) red, \(5\) green and \(6\) yellow counters.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. The first counter is replaced, so the probabilities are the same for Niklas and Sasha.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. P(same colour)=
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. \left(frac4{15}\right)^2+
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. \left(frac5{15}\right)^2+
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. \left(frac6{15}\right)^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. =frac{16+25+36}{225}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. =frac{77}{225}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. Answer: \(frac{77}{225}\).
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: frac{77}{225}.

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X.
Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

Answer reference | May-Nov 2020 | Question 20

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

20	$\dfrac{\left(\dfrac{X+4}{2}\right)}{X}\left(=\dfrac{X+4}{2X}\right)$ **or** $\dfrac{\left(\dfrac{X+4}{2}\right)-1}{X-1}\left(=\dfrac{X+2}{2X-2}\right)$	eg, where b = number of blue counters $\dfrac{b}{2b-4}$ **or** $\dfrac{b-1}{2b-5}$	eg, where r = number of red counters $\dfrac{r+4}{2r+4}$ **or** $\dfrac{r+3}{2r+3}$			M1	for making a correct start by finding the probability of the first counter being blue for their method
	eg $\dfrac{X+4}{2X}\times\dfrac{X+2}{2X-2}$	eg $\dfrac{b}{2b-4}\times\dfrac{b-1}{2b-5}$	eg $\dfrac{r+4}{2r+4}\times\dfrac{r+3}{2r+3}$			M1	oe correct calculation for 2 blue (using one variable)
	eg $8(X^2+6X+8)=3(4X^2-4X)$	eg $8b(b-1)=3(2b-4)(2b-5)$	eg $8(r+4)(r+3)=3(2r+4)(2r+3)$			M1	dep for a correct equation with no algebraic fractions eg could have $X^2+6X+8=\dfrac{3}{8}(4X^2-4X)$
	Eg $4X^2-60X-64\ (=0)$ **or** $X^2-15X-16\ (=0)$ oe	eg $4b^2-46b+60\ (=0)$ **or** $2b^2-23b+30\ (=0)$ oe	eg $4r^2-14r-60\ (=0)$ **or** $2r^2-7r-30\ (=0)$ oe			M1	for rearranging their equation to a correct 3 term quadratic
				16	5	A1	cao dep on M4
							Total 5 marks

Worked solution

May-Nov 2020 | Question 20 | 5 marks

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

Official final mark-scheme pages are embedded immediately after this question.